

Disclosure in Incentivized Reviews: Does It Protect Consumers?

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Abstract. The well-documented rating inflation of incentivized reviews (IRs) can mislead consumers into choosing a product that they would otherwise not buy. To protect consumers from this undesirable influence, the U.S. Federal Trade Commission recommends that reviewers conspicuously disclose any material connection they may have with sellers. In theory, such disclosures safeguard consumers by motivating reviewers to be truthful and inducing consumers to discount inflated IR ratings. Our research finds, however, that IR disclosure accomplishes neither. Specifically, our empirical analysis of consumer reviews on Amazon reveals that, even with disclosure, (1) rating inflation of IRs remains, and (2) this inflation boosts sales at consumers' expense. Finally, we propose an alternative approach to eliminate rating inflation of IRs and empirically demonstrate its effectiveness. These findings have important implications for consumers, firms, and ongoing policy discussions around IRs.

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Online shopping runs on reviews... If 500 other people have bought something and say it works, you can have a lot more confidence. But what if those people were paid to leave those positive reviews?¹

—Lina M. Khan, chair, Federal Trade Commission

1. Introduction

Chair Khan's statement refers to the pressing problem of incentivized reviews (IRs), whereby a firm provides a material incentive (e.g., free product, bonus, or cash) to motivate consumers to post online reviews of its products. Although such incentives increase the volume of a product's user-generated information (Shi and Wojnicki 2014, Burtch et al. 2018), research shows that they markedly increase the favorability of the reviews that the product receives (e.g., Cabral and Li 2015, Khernamnuai et al. 2018). This rating inflation puts shoppers

at risk for selecting products they otherwise would not have bought. Because online product reviews have become a key channel for consumers to gather product information and IRs are so prevalent, the social welfare implications of IR bias can be substantial.

In the United States, the Federal Trade Commission (FTC) is charged with discouraging deceptive trade practices. To address the problem of bias in IRs, the FTC recently issued guidance that IRs be accompanied by a conspicuous disclosure of the material relationship between the reviewer and the seller (see, for example, the FTC's "Featuring Online Customer Reviews: A Guide for Platforms," January 2022).² The FTC's recommendation that IRs incorporate disclosure has been embraced by prominent advertisers³ and product review platforms.⁴

Intuitively, incorporating disclosure into incentivized reviews has the potential to help consumers in two

ways. First, from the reviewer's perspective, because public disclosure draws a spotlight on the reviewer's conflict of interest, the reviewer might be motivated to be more truthful (Snyder 1974). This effect could be especially strong for reviewers who care about their reputation (Pinch and Kesler 2011). Second, from the consumer's perspective, disclosure serves an information function, that is, alerting shoppers that a particular review may be biased and inducing them to adjust their assessment of it.

As regulators and market stakeholders build momentum for instituting disclosure as the solution to IR bias, neither the ability of disclosure to curb incentivized reviewers' rating inflation nor its effectiveness in helping consumers discount such inflation has been carefully examined in real markets. This paper empirically investigates the effectiveness of IR disclosure on reviewers and consumers. We also propose and test an alternative approach to IRs that eliminates rating inflation by avoiding the underlying conflict of interest.

We begin by examining whether IR disclosure can effectively eliminate rating inflation. Specifically, we compare reviews with an incentive disclaimer (i.e., disclosed IRs or DIRs) to reviews with a verified purchase tag (i.e., verified reviews or VRs) on Amazon. After controlling for review- and reviewer-specific characteristics, we find that, for the same product, DIRs have higher average ratings than do VRs. Note that some VRs in our data can be undisclosed IRs with more favorable ratings (i.e., written by reviewers who received an incentive but failed to disclose their conflict of interest). This suggests that the average rating of organic reviews written by users with no financial incentive may be even lower than that of VRs, implying that the rating inflation of DIRs can be even greater than we report. More troublingly, we also find that the rating difference between DIRs and VRs is larger for low- than for high-quality products.

Our finding that ratings associated with DIRs exceed those of VRs is important because it shows that having IR disclosure in place is insufficient to eliminate the rating inflation of IRs. Our finding that DIR rating inflation for low-quality products is higher than for high-quality products is especially bad news for consumers because more harm is caused when an inflated rating induces the purchase of a low-quality product.

Given the inability of disclosure to eliminate IR rating inflation, it is crucial to examine whether disclosure can serve an effective information function to prevent consumers from being misled by rating bias. To do so, we exploit an abrupt policy change made by Amazon on October 3, 2016. For several years prior to this date, Amazon allowed IRs if reviewers disclosed their financial incentive. After this date, however, Amazon prohibited the posting of new incentivized reviews and gradually deleted existing ones. This policy change provides an opportunity to compare the prepolicy and

postpolicy periods for the same product and estimate the effect of DIRs (or, more precisely, the effect of their removal).

Our difference-in-differences (DID) estimates, based on Amazon data surrounding the date of its policy change, lead to three important findings. First, DIRs have a positive sales effect, suggesting that, when incentivized reviewers disclose their conflict of interest, the inflated reviews can still effectively induce more consumers to buy a product. Second, such a positive sales effect is stronger for products with higher IR rating inflation, confirming that, even with disclosure, rating inflation directly contributes to the sales increase induced by IRs. Third, consumers are more likely to give poor postpurchase product evaluations (e.g., one-star ratings) in the presence than in the absence of DIRs, suggesting that, even with disclosure, IRs lower consumers' postpurchase product satisfaction. Taken together, these findings provide evidence that IR disclosure is not effective in helping consumers discount IR rating inflation. Despite disclosure, inflated ratings continue to enhance sales and consumers remain vulnerable to a disappointing consumption experience. We replicate these findings via an alternative cross-platform DID design, employing data from two platforms that implemented the same policy change at two different points in time: Amazon US and Amazon UK.

Finally, given that disclosure neither eliminates DIR rating inflation nor enables consumers to effectively discount inflated ratings, we propose and test a different way to remove rating inflation of IRs. Based on incentive compatibility between the two parties engaged in the creation of an IR, that is, the firm and the reviewer, we conjecture that offering rewards to reviewers leads to inflated ratings when IRs are commissioned by a firm that benefits directly from sales of the reviewed products. Such bias can be removed, however, if IRs are instead commissioned by an independent platform (that benefits from sales of many products from various suppliers). We empirically test our conjecture by formally defining two types of IR, (a) firm-initiated and (b) platform-initiated (PIRs), which differ only in who commissions them. Examples of the latter include Amazon's Vine program and Home Depot's Seeds program, in which the platform distributes many brands of free products to reviewers. We test our conjecture using Amazon's Vine program, under which IRs are disclosed with a badge, that is, "Vine Customer Review of Free Product." Results of our empirical examination of Amazon's Vine program show that, unlike firm-initiated DIRs, we find no evidence that ratings of platform-initiated DIRs are systematically different from those of VRs. These results suggest that moving from a firm- to a platform-initiated IR system is an effective way to reduce IR rating inflation. This provides empirical evidence of the advantage of a platform-initiated IR system.

Our research contributes by advancing the academic literature, alerting consumers, and offering insights for public policy. First, as an important societal concern, IRs have garnered increasing academic attention in recent years. Some studies adopt a reviewer's perspective, investigating how incentives influence review characteristics (such as rating, sentiment, length, and quality) and reviewers' future review participation (e.g., Cabral and Li 2015, Burtch et al. 2018, Khern-am-nuai et al. 2018, Qiao et al. 2020). Other studies take the firm's perspective, exploring the sales effect of IRs and the characteristics of firms that sponsor them (e.g., Li et al. 2020, Fradkin and Holtz 2023). Still other studies use game theory to model incentives, identify conditions that promote IR bias, and explore the impact of IR bias on social welfare (Dellarocas 2006, Mostagir and Siderius 2023).⁵ Our work enhances the literature on IRs by focusing on the effectiveness of IR *disclosure*, which is a commonly prescribed regulatory remedy for IR bias.

Second, rating inflation by IRs is a problem to which all online shoppers can relate. Our analysis directly benefits consumers by drawing their attention to IR bias, the failure of disclosure to counter that bias, and the benefits of platform-initiated IRs.

Finally, our investigation is directly relevant to the FTC's active rulemaking process on the regulation of reviews and endorsements, including IRs.⁶ Our research provides compelling empirical evidence that disclosure neither eliminates IR rating inflation nor prevents consumer harm. Further, we propose and empirically demonstrate that moving from a firm- to a platform-initiated IR system can eliminate rating inflation by removing the conflict of interest underlying it. We note that it might be more beneficial to consumers if policymakers foster the growth of platform-initiated review systems rather than nudging consumers to pay attention to disclosures embedded in firm-initiated IRs.

2. Empirical Context

2.1. Background

Our study examines IRs on Amazon, which, as a platform provider, has taken several proactive measures to keep reviews informative and unbiased. In September 2009, Amazon started displaying a verified purchase tag on reviews from purchases made through its own site (Anderson and Simester 2014). The verified purchase tag provides consumers with information that helps them decide which reviews are more trustworthy. More recently, Amazon announced a more stringent policy: from October 3, 2016, onward, Amazon banned IRs and subsequently allowed no newly posted IRs. Moreover, since this announcement, Amazon has deleted all existing IRs from its website.⁷

In this context, we examine two types of reviews. First, DIRs are reviews with a disclaimer (of the material

connection) in the text. Note that DIRs are truly incentivized because there is no reason for a reviewer to write a disclaimer without having a material connection with a firm.⁸ Second, as a benchmark, we also examine VRs, that is, reviews with a verified purchase tag. By definition, VRs are posted by consumers who actually purchased the product on Amazon. While we expect that most VRs are authentically produced, it is possible that some might have been incentivized by firms. We classify a VR as a DIR whenever it contains a disclaimer, but we cannot completely rule out the existence of undisclosed incentivized reviews among VRs. To the extent that we distinguish between the two types of reviews, however, their existence strengthens our results: when we compare DIRs with VRs (in Section 3), the comparison becomes more conservative.

2.2. Data

We combine two sources of data: (1) sales and (2) review data. We collect the former from the website [keepra.com](https://www.keepra.com) and the latter directly from Amazon. First, our sales data collection largely follows the literature (e.g., Park et al. 2023). Specifically, because sales data on Amazon are not publicly available, we collect sales rank data to proxy sales. This approach is well-substantiated in the literature (e.g., Chevalier and Goolsbee 2003, Sun 2012). Moreover, the sales rank serves our purposes especially well because, in our study of the sales impact, we are interested in the change rather than the absolute level of sales.

Second, our review data were collected twice. In September 2016, we collected from the Amazon website all reviews of 5,273 products for which at least one DIR was posted by top 1,000 ranked reviewers (see Section A1 of the online appendix for details). This review data set contains 654,463 reviews, including 241,831 DIRs. After Amazon's announcement of its IR policy change (October 2016), we again collected all reviews for the same set of products, specifically in March 2017. For our analysis in Section 3, we use reviews on all products in the first data set that were not affected by Amazon's policy change. For the analysis of product sales in Section 4, however, we use only the matched products in both the review and the sales data sets. Because [keepra.com](https://www.keepra.com) only keeps track of products above a certain sales level, we observe 4,147 matched products.

Besides our main data set, we construct additional data sets for various analyses in the paper. The detailed construction of all the data sets used in the paper is reported in Online Section A1. Table 1 lists these data sets along with their key construction procedures and the analyses in which they are used.

3. Impact of IR Disclosure on Reviewers

In this section, we examine whether the disclosure requirement for IRs can change incentivized reviewers'

Table 1. Data Sets

Data set	Construction	Number of products	Analysis
Main	• Products with at least one DIR • Reviews: Before and after DIR removal	• Before Amazon US policy change: 5,273 • After Amazon US policy change: 5,101	• Difference in ratings: DIRs versus VRs (Section 3) • The impact of DIR removal on sales (Section 4) • The impact of DIR removal on the number of one-star VRs (Section 4)
Consumer Reports	• Products with quality scores from <i>Consumer Reports</i> in appliances and electronics	• 1,723 with quality score • 1,616 (among 1,723) had at least one review from Amazon	• Difference in ratings between DIRs and VRs: moderation by quality (Section 3) • Difference in ratings: Vine reviews versus VRs (Section 5) • Difference in ratings between Vine reviews and VRs: moderation by quality (Section 5)
Amazon UK	• A subset of products in the main data set matched with products from Amazon UK	• 616 matched products with at least one review from Amazon UK	• The impact of DIR removal on sales: cross-platform analysis (Section 4) • The impact of DIR removal on the number of one-star VRs: cross-platform analysis (Section 4)
Top reviewers	• All reviews posted by Amazon's top 1,000 ranked reviewers as of April 2016	• 708,091 products with at least one review from top 1,000 ranked reviewers	• Difference in ratings between DIRs and VRs: within-reviewer analysis (Section 3)

Note. In the main data set, 172 products became unavailable when we collected the review data set after the policy change.

posting behavior. If the disclosure requirement disciplines reviewers to be impartial, we expect no difference in the ratings of DIRs from those of VRs. However, different rating levels imply that the requirement is not sufficient to curb the incentive to distort reviews. Thus, we investigate how DIRs fare against VRs in terms of their ratings.

To begin, a simple mean comparison shows that the average rating of DIRs is substantially higher than that of VRs (4.73 versus 4.08; $p < 0.01$). Moreover, as reported in Table 2, most DIRs are positive with 95.7% of them exhibiting ratings of four or five (compared with 75.3% of VRs). However, this comparison may be driven by the characteristics of products, reviewers, or posting timing of DIRs that are distinct from those of VRs. Thus, we formally test the ratings difference by considering the following regression of individual ratings of both DIRs and VRs with a set of control variables:

$$Rating_{ij} = \alpha \cdot DIR_{ij} + \beta X_{ij} + \gamma_i + \varepsilon_{ij}, \quad (1)$$

where $Rating_{ij}$ is the rating of review j for product i , DIR_{ij} is a dummy variable indicating review j is a DIR (as opposed to a VR), X_{ij} is a vector of controls, γ_i are product-level fixed effects, and ε_{it} is the error term. Note that, because we absorb cross-product variation with γ_i , our estimation only utilizes within-product variation in ratings. Moreover, by including a set of individual review-specific controls (X_{ij}), we account for reviewer-as well as review-specific characteristics. Specifically, we include reviewer ranking and the number of reviews written by the authoring reviewer to account for

reviewer heterogeneity as well as *Time* (i.e., the number of days since the first review was posted) and *Order* (i.e., the position of the review in the sequence of reviews of the product) of the review to account for temporal and sequential dynamics (Godes and Silva 2012). The inclusion of product-level fixed effects enables us to control for both observed and unobserved time-invariant product characteristics. As noted earlier, to fully utilize our sample of DIRs, we employ data collected prior to the Amazon policy change (i.e., September 2016) for which we restrict our attention to products with at least one DIR and one VR. Finally, we cluster the standard errors at the product level to account for correlation patterns in errors (Bertrand et al. 2004).

In Equation (1), the coefficient of our primary interest is α , which captures the ratings difference between DIRs and VRs. As reported in column (1) of Table 3, we find the estimate of α is positive and highly significant

Table 2. Rating Distributions: DIRs and VRs

Rating	DIRs	Verified reviews
Rating: 1, %	0.49	11.11
Rating: 2, %	0.84	5.89
Rating: 3, %	2.94	7.75
Rating: 4, %	17.09	14.58
Rating: 5, %	78.64	60.67
Average ratings	4.73	4.08
N	241,831	310,580

Note. For this table, we use DIRs and VRs posted to 5,273 products in the main data set collected before the policy change.

Table 3. Difference in Ratings: DIRs and VRs

	(1)		(2)	
	Estimate	Standard error	Estimate	Standard error
DIR	0.480***	0.009	0.486***	0.086
DIR × Quality			−0.367**	0.140
Log reviewer ranking	4.23E-04	1.26E-03	9.64E-03	1.05E-02
Log number of reviews of reviewer	0.071***	0.003	0.102***	0.013
Time	−1.41E-04***	2.70E-05	−1.63E-04***	5.27E-05
Order	−2.95E-04***	6.00E-05	2.15E-07	6.77E-07
Product fixed effects	Yes		Yes	
R ²	0.204		0.043	
N	544,014		389,175	

Notes. Standard errors are clustered at the product level. The unit of analysis is an individual review. For column (1), we use DIRs and VRs posted to a subset of products in the *Main* data set. Specifically, we focus on 5,191 products with at least one DIR and one VR to exploit the within-product variation. For column (2), we use DIRs and VRs posted to products in the *Consumer Reports* data set. Note that all 105 products used in column (2) have quality scores, and each product received at least one DIR and one VR.

Significantly different from zero at $p < 0.05$; *significantly different from zero at $p < 0.01$.

($\hat{\alpha} = 0.48$, $p < 0.01$), suggesting that DIRs are, on average, 0.48 stars higher than VRs for the same product. This substantial difference in ratings suggests that the disclosure requirement does not provide a sufficient incentive for incentivized reviewers to remove rating inflation from their reviews. This finding can be attributed to psychological mechanisms described in behavioral research. For example, reviewers may exhibit both moral licensing and strategic exaggeration (Cain et al. 2005, Loewenstein et al. 2011). Under moral licensing, a reviewer's disclosure may relieve the reviewer of guilt over writing a biased review. Under strategic exaggeration, a reviewer may consciously increase the bias in the review to counteract any discounting that disclosure may cause.

In the data set used to estimate Equation (1), 98% of the reviews were written by different reviewers. This raises the possibility that the estimated rating difference might be driven by reviewer selection. Whereas we have already controlled for the reviewer-specific characteristics, we can rule out such a possibility by conducting an additional within-reviewer analysis. Specifically, among Amazon's top 1,000 ranked reviewers, we identify 620 reviewers who have written one or more reviews of both types (i.e., DIR and VR) and compare their ratings of DIRs and VRs, incorporating reviewer-level fixed effects and clustering the standard errors at the reviewer level. We find that, even within the same reviewer, the ratings for DIRs consistently appear higher than those for VRs by 0.17 stars ($p < 0.01$). This reaffirms the inability of the disclosure to curb the distortion incentive of reviewers (for details of this analysis, see Online Section A2).

Finally, one may wonder whether the rating inflation of DIRs may be more pronounced for products of a certain quality. To answer this question, we constructed the *Consumer Reports* data set by collecting quality scores for all products in 48 categories of appliances and electronics in *Consumer Reports*. For each product, we also collect

review data from Amazon, but because we use within-product variation in this analysis, we focus on 105 products that received one or more reviews of both types (DIRs and VRs) on Amazon at the time of data collection (i.e., August 2016).⁹

We estimate Equation (1) by including an interaction term $IR_{ij} \times Q_i$, where Q_i is a percentile of product i 's quality score within a category. We report the estimation result in column (2) of Table 3. Results show that the rating inflation of DIRs *decreases* as the product's quality *increases*. In other words, DIR rating inflation is systematically higher for lower quality products.¹⁰ Greater rating distortion at the lower end is particularly bad for consumers because more harm is caused when an inflated rating induces the purchase of a low-quality product. Taken together, our findings suggest that a mere disclosure requirement may not prevent information distortion by IRs.

4. Impact of IR Disclosure on Consumers

In the previous section, we show that the disclosure requirement is not sufficient to remove reviewers' incentives to distort their reviews. It remains unclear, however, whether purchase decisions are influenced by such inflated ratings because consumers may discount the information contained in DIRs when they see a disclosure. In this section, we examine whether disclosure helps consumers sufficiently discount rating inflation by investigating whether DIRs may fail to alter product sales. In the case of sales increase, we also seek evidence that consumers are misled by rating inflation.

To estimate the sales impact of DIRs, we exploit the fact that Amazon did not delete all IRs immediately after the policy change in October 2016. Specifically, we observe that 35% of DIRs in our data set had not been removed six months after the policy change (March 2017). This implies that there are two groups of products

in our data: one group for which some (or all) DIRs had been removed and another group for which none had yet been removed.¹¹ By comparing these two groups, we can identify the impact of DIRs or, precisely, the impact of DIR removal. More specifically, the first group of products (having lost some DIRs) serves as our treatment group, whereas the second group (having lost no DIRs) comprises the control group. We estimate the effect of DIR removal by measuring the temporal change in the sales of the treatment group relative to that of the control group, via a DID approach.

We consider the following difference-in-differences specification, using preperiod data from April to September 2016 and postperiod data from October 2016 to March 2017 with the unit of analysis being product (i) $\times$ day (t):

$$\ln(S_{it}) = \lambda \cdot (Treat_i \times Post_t) + \beta X_{it} + \gamma_i + M_t + \varepsilon_{it}, \quad (2)$$

where $\ln(S_{it})$ is the log sales rank for product i on day t , $Treat_i$ is a binary indicator for products in the treatment group, $Post_t$ is a binary indicator for observations after the policy change, X_{it} is the vector of control variables, γ_i are product-level fixed effects, M_t are year-month fixed effects, and ε_{it} is the error term. As controls, we include average ratings, log number of reviews, log price, and the category-specific log product age (the number of days posted on the platform).¹² Note that the inclusion of product-level fixed effects not only controls for product characteristics, but also helps us to reduce the concern for potential selection in the group assignment. As shown in Online Section A4, the group assignment can be predicted by the number of preexisting DIRs, which, if controlled, makes the group assignment largely idiosyncratic to the dependent variable. In our formulation, product-level fixed effects account for the number of preexisting DIRs. Finally, to account for correlation in the errors, we cluster standard errors at the product level (Bertrand et al. 2004).

An important identifying assumption of our DID approach is that time trends in the outcome variable would have been similar for both groups if DIRs had not been removed. To explore the validity of this assumption, we estimate the month-specific group difference on Equation (2) by replacing $Post_t$ with year-month dummies. As shown in Figure 1(a), we find no evidence of differential trends for the outcome variable before the policy change, thus lending support for our identifying assumption.

Given this, we discuss our estimation result. Under this specification, the product-level fixed effects capture the underlying difference in $\ln(S_{it})$ across products, and the year-month fixed effects capture the common time trends in $\ln(S_{it})$ before and after the policy change. Then, the coefficient of $Treat_i \times Post_t$ captures the difference in changes in the log sales rank before and after the policy

change between treatment and control groups. As reported in column (1) of Table 4, the estimated coefficient of $Treat_i \times Post_t$ is positive and highly significant ($\hat{\lambda} = 0.087$, $p < 0.01$). This result shows that, after the policy change, the sales rank of products in the treatment group increased 9.1% ($= e^{0.087} - 1$) more than those in the control group, indicating that the removal of DIRs has a negative impact on sales given the inverse relationship between sales and sales rank. Conversely, this result suggests that the presence of DIRs increases product sales, thus establishing a positive impact of DIRs on sales.

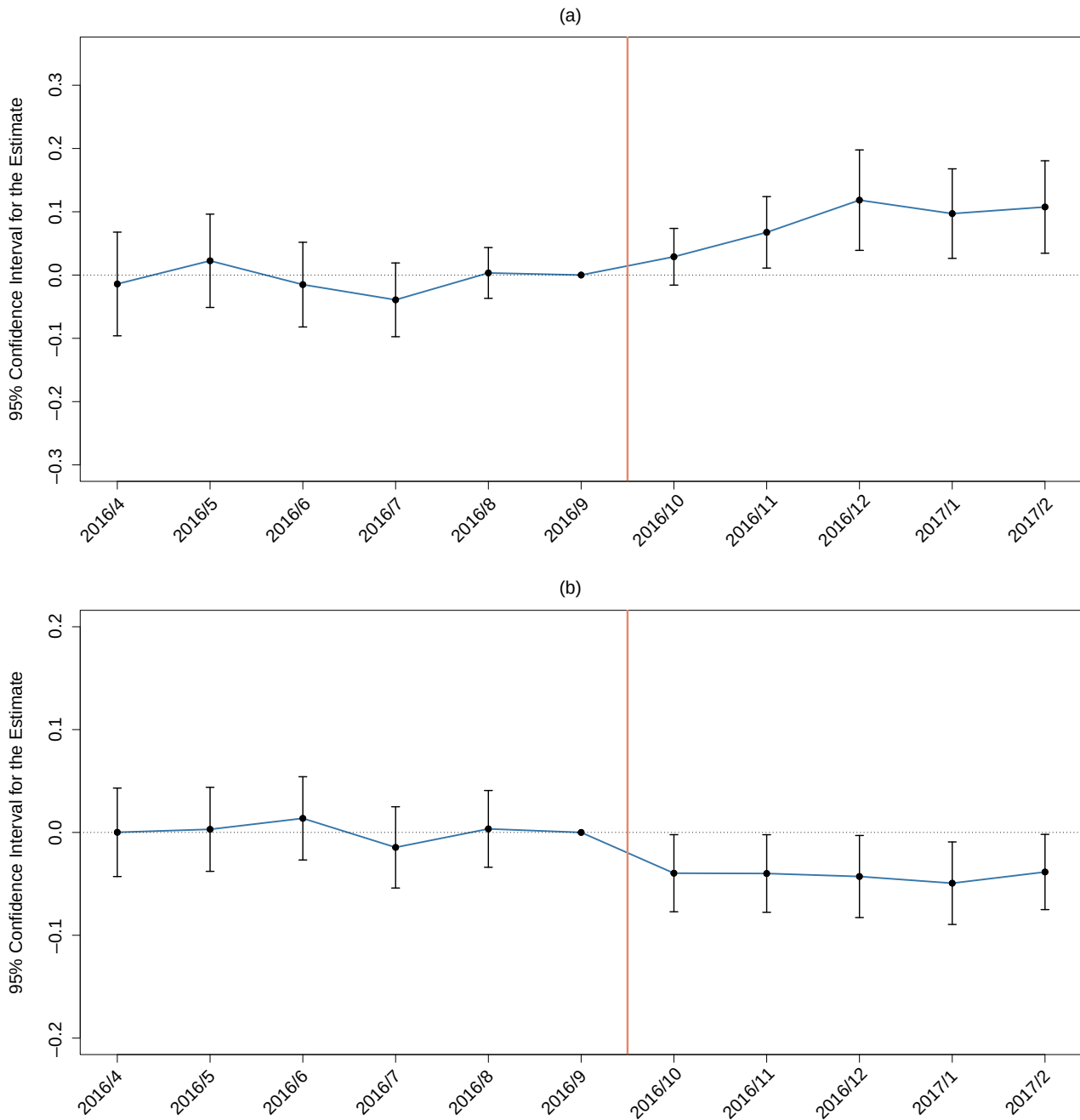
Notably, the sales increase occurs even when the material connection between firms and reviewers is disclosed, implying that disclosure does not motivate consumers to fully discount DIR rating inflation. Psychological mechanisms that could hamper consumers' response to disclosure include their failure to notice the disclaimers because of information overload, inability to fully discount the biased advice (Tversky and Kahneman 1974), and/or increased trust of reviewers if the disclosure is interpreted as a sign of honesty (Pearson et al. 2006).

Given this result, we next explore whether inflated ratings of DIRs indeed contribute to the estimated sales change. For this purpose, we operationalize the rating inflation as the difference in the average ratings with and without DIR removal and denote it by ΔR . Figure 2 shows that, in our data, ΔR is heterogeneous across products in the treatment group. Thus, we divide the treatment group into two groups using a median split and examine whether the DIR removal effect is greater in the high ΔR group (i.e., products that experienced a larger reduction in average ratings). Specifically, we separately estimate Equation (2) for the two groups, keeping the control group the same, and compare their estimated effects.

We report our estimation results in Table 4. The estimate of λ is 0.078 ($p < 0.05$) for the low- ΔR group and 0.109 ($p < 0.01$) for the high- ΔR group (columns (2) and (3)) with the estimate being larger on the high- ΔR group. To formally test whether the high- ΔR group experienced a greater reduction in sales compared with the low- ΔR group after the policy change, we reestimate Equation (2) using the low- ΔR group as the control group and the high- ΔR group as the treatment group. In this within treatment group approach, we find that the estimated λ is positive and significant ($\hat{\lambda} = 0.051$, $p < 0.05$), suggesting that the sales of the high- ΔR group decreased significantly more than those of the low- ΔR group (column (4)). These results confirm that, even with disclosure, rating inflation directly contributes to the sales change induced by DIR removal.

Our findings thus far suggest that DIRs increase product sales by misleading consumers through their inflated ratings. To provide direct evidence that consumers are

Figure 1. (Color online) Impact of the Removal of DIRs



Notes. In this figure, we report the month-specific estimates of λ along with 95% confidence intervals. (a) Log sales rank. (b) Log number of one-star verified reviews.

misled by inflated ratings of DIRs, we look into the number of one-star VRs. If DIRs had misled some consumers and made them unsatisfied, DIR removal should decrease the number of one-star VRs. We specifically estimate the following equation:

$$\ln(1 + N_{it}^{One\ star}) = \lambda \cdot (Treat_i \times Post_t) + \gamma_i + \mathbf{M}_t + \varepsilon_{it}, \quad (3)$$

where $N_{it}^{One\ star}$ is the number of VRs with one-star

ratings posted for product i in month T . First, as shown in Figure 1(b), we find no evidence of differential trends in the outcome variable between the two groups before the policy change. Given this, we find that the estimate of λ is negative and significant ($\hat{\lambda} = -0.043$, $p < 0.01$), suggesting that the number of one-star VRs decreased by 4.3% because of the DIR removal (column (1) of Table 5).¹³ We interpret this finding as evidence that inflated ratings of DIRs misled some consumers into

Table 4. The Impact of the DIR Removal on Sales

	(1)		(2)		(3)		(4)	
	Full sample		Low- ΔR group		High- ΔR group		Within-treatment group	
	Estimate	Standard error	Estimate	Standard error	Estimate	Standard error	Estimate	Standard error
Treat $\times$ Post	0.087***	0.027	0.078**	0.031	0.109***	0.032	0.051**	0.023
Log number of reviews	−0.238***	0.030	−0.265***	0.041	−0.218***	0.040	−0.220***	0.031
Average ratings	−0.473***	0.054	−0.468***	0.086	−0.471***	0.066	−0.438***	0.056
Log price	0.519***	0.031	0.525***	0.045	0.486***	0.040	0.541***	0.031
Product fixed effects	Yes		Yes		Yes		Yes	
Year-month fixed effects	Yes		Yes		Yes		Yes	
R^2	0.818		0.829		0.816		0.812	
N	982,999		537,231		574,336		854,431	

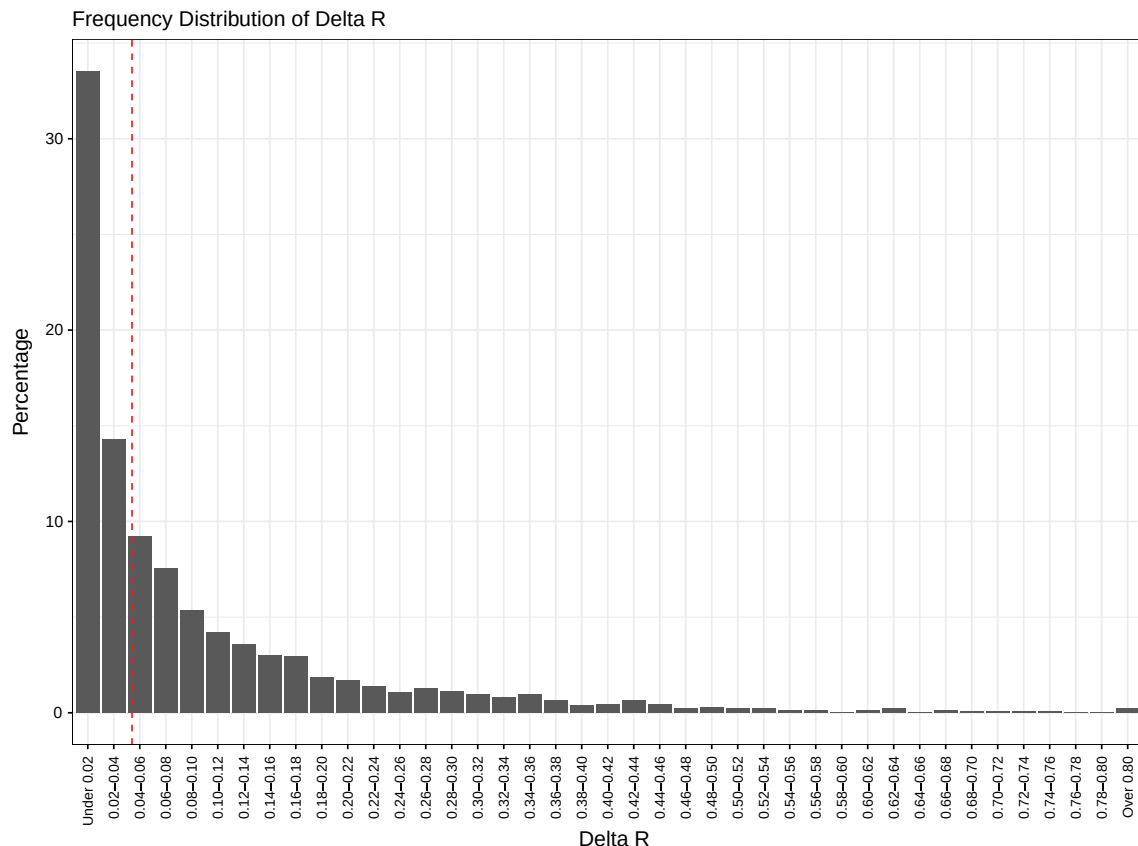
Notes. The unit of analysis is Product $\times$ Day. The Standard errors are clustered at the product level. The log number of reviews and the average ratings are computed based on all reviews but DIRs. In columns (2) and (3), we use the low- ΔR and high- ΔR groups as the treatment groups, respectively, while keeping the control group the same. In column (4), the low- ΔR group serves as the control group, and the high- ΔR group serves as the treatment group.

Significantly different from zero at $p < 0.05$; *significantly different from zero at $p < 0.01$.

making unsatisfactory purchases. In Online Section A5, moreover, we show that the magnitude of this effect is higher for the high- than the low- ΔR group, implying that DIRs with greater rating inflation are even more misleading to consumers. These results provide further evidence that the disclosure in IRs does not help consumers sufficiently discount their rating inflation.

4.1. Robustness Checks

So far, we identify the effect of DIRs on sales by comparing products from which DIRs were removed with those whose DIRs were retained in the first six months after the policy change. We now show the robustness of our results by adopting a different design, which compares the sales of the same products across two

Figure 2. (Color online) Distribution of ΔR 

Note. A vertical dotted line represents the median.

Table 5. The Impact of the DIR Removal on the Number of One-Star Verified Reviews

	(1)		(2) Cross-platform analysis	
	Estimate	Standard error	Estimate	Standard error
Treat × Post	−0.043***	0.012	−0.073***	0.022
Product fixed effects	Yes			
Product-platform fixed effects			Yes	
Year-month fixed effects	Yes		Yes	
R ²	0.402		0.411	
N	44,858		4,016	

Notes. In column (1), the unit of analysis is a Product × Month. Standard errors are clustered at the product level. We focus on products that received at least one VR during the analysis period. In column (2), the unit of analysis is a Product × Platform × Month. Standard errors are clustered at the Product × Platform level. We use products that received at least one VR on both Amazon US and Amazon UK during the analysis period.

***Significantly different from zero at $p < 0.01$.

platforms, one with a policy change and the other with no change. This alternative design exploits the fact that Amazon UK did not change its IR policy until November 22, 2016.¹⁴ for the policy announcement date on Amazon UK). Thus, focusing on the period before November 22, 2016, we identify the sales impact of DIR removal by the cross-platform difference in the sales change of the same products before and after the Amazon US policy change date. Note that this design is not subject to the potential selection discussed earlier on our main design because we now compare the sales of the same products on the two platforms.

For the analysis, we construct the Amazon UK data set. We specifically collect review and sales data on a subset of products from the *Main* data set that are also available from Amazon UK.¹⁵ Using observations prior to November 22, 2016, we estimate the following equation:

$$\ln(S_{it}^k) = \lambda \cdot (\text{Treat}^k \times \text{Post}_t) + \beta \mathbf{X}_{it}^k + \gamma_i^k + \mathbf{M}_t + \varepsilon_{it}^k, \quad (4)$$

where $\ln(S_{it}^k)$ is the log sales rank for product i on date t and platform $k \in \{\text{US}, \text{UK}\}$; Treat^k is an indicator for the platform with DIR removal, taking the value of one for Amazon US and zero for Amazon UK; Post_t is the indicator for observations after the policy change (i.e., one for October 3 to November 22, 2016, and zero for April 1 to October 2, 2016); $\mathbf{X}_{it}^k$ is the vector of the same set of controls used in Equation (2); and γ_i^k are product–platform specific fixed effects. Note that γ_i^k absorb any potentially differential consumer preferences across the two platforms for the same product. Finally, standard errors are clustered at the product–platform level.

The coefficient of our primary interest is λ , which captures the sales impact of DIR removal from the Amazon US platform. Consistent with our main result, we find that the estimate of λ is positive and significant ($\hat{\lambda} = 0.230$, $p < 0.05$), suggesting that products on Amazon US experienced a measurable sales reduction after

the policy change compared with the same products on Amazon UK, which experienced no policy change (Table 6). As in our main model, we check the validity of the parallel-trends assumption by replacing Post_t with year-month dummies in Equation (4). As shown in panel (a) of Figure 3, we find no evidence of differential trends in the outcome variable before the policy change.

Similarly, we check for the robustness of the impact of DIRs on the number of one-star verified reviews using the cross-platform design. We specifically estimate the following equation:

$$\ln(1 + N_{iT}^{\text{One star}, k}) = \lambda \cdot (\text{Treat}^k \times \text{Post}_t) + \gamma_i^k + \mathbf{M}_t + \varepsilon_{it}^k. \quad (5)$$

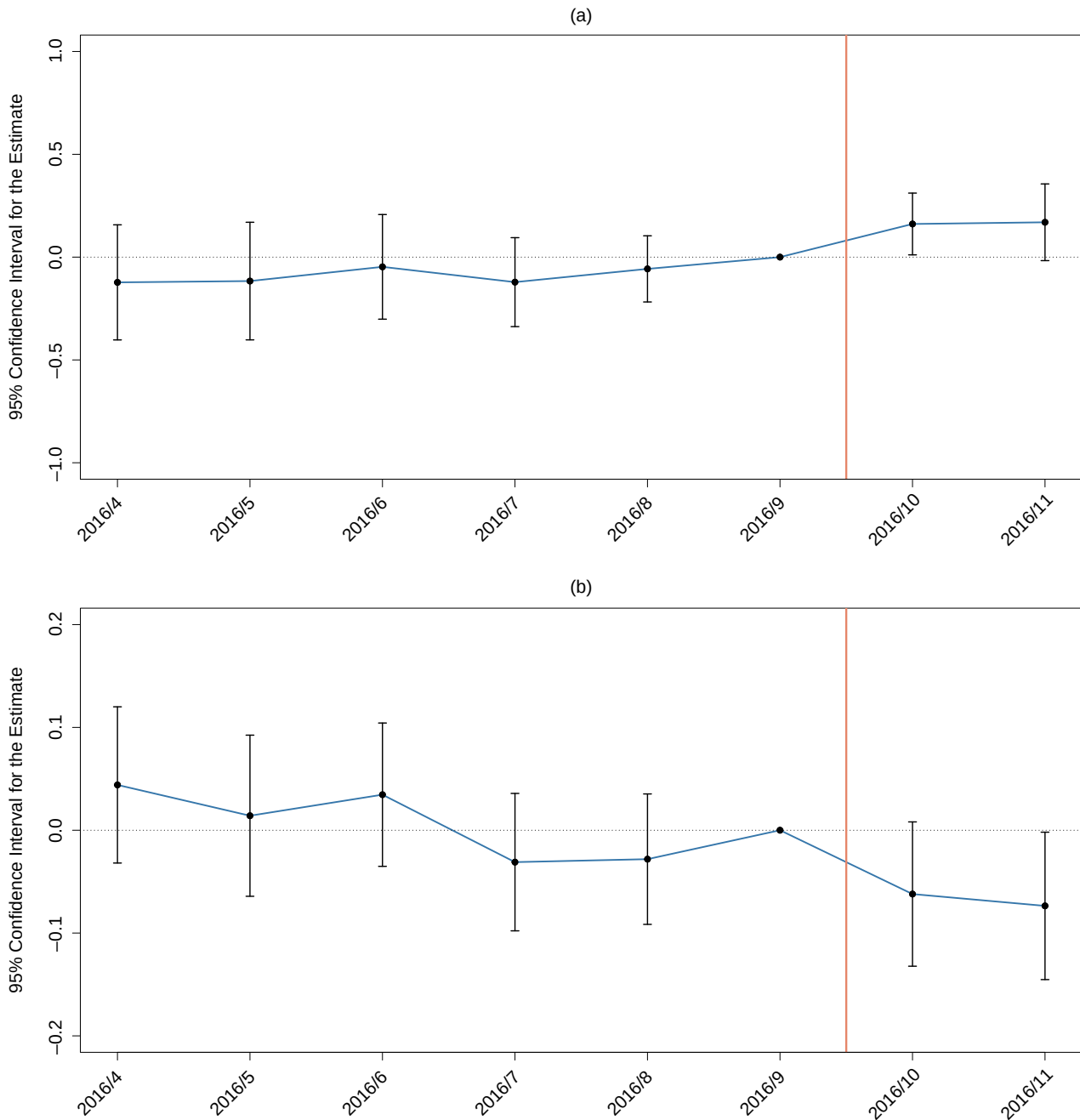
Consistent with our main result as reported in column (2) of Table 5, we find that the estimate for λ is negative

Table 6. The Impact of the DIR Removal on Sales: Cross-Platform Analysis

	Estimate	Standard error
Treat × Post	0.230**	0.093
Log number of reviews	0.046	0.092
Average ratings	0.065	0.144
No-review dummy	0.654	0.679
Log price	0.747***	0.109
Product-platform fixed effects	Yes	
Year-month fixed effects	Yes	
R ²	0.832	
N	33,550	

Notes. The unit of analysis is Product × Platform × Day. Standard errors are clustered at the Product × Platform level. The log number of reviews and the average ratings are computed based on all reviews other than DIRs. The average rating is coded as zero when there is no review. In this table, we restrict our attention to products that lost at least one DIR from Amazon US to measure the DIR-removal effect. We use products for which sales rank and price data are available from Keepa on both Amazon US and Amazon UK during the analysis period.

Significantly different from zero at $p < 0.05$; *significantly different from zero at $p < 0.01$.

Figure 3. (Color online) Impact of the Removal of DIRs: Amazon US – Amazon UK Design

Notes. In this figure, we report the month-specific estimates of λ along with 95% confidence intervals. (a) Log sales rank. (b) Log number of one-star verified reviews.

and significant ($\hat{\lambda} = -0.072$, $p < 0.01$). This implies that the removal of DIRs led to a decrease in the number of one-star verified reviews on Amazon US in comparison with Amazon UK from which DIRs were not removed. Note that we find no evidence of differential trends in the outcome variable before the policy change as shown in panel (b) of Figure 3.

Finally, our results are robust to alternative samples. We specifically show the robustness on (1) a common support subsample (to rule out the effect of group

difference) and (2) a subsample of products losing only DIRs after the policy change (to rule out the effect of undisclosed IR removal). We provide details of these robustness checks in Online Section A6.

5. An Alternative Solution: Platform-Initiated IRs

Our empirical results so far reveal that the disclosure neither reverses reviewers' incentive to distort their IR

ratings nor motivates consumers to fully discount the ratings inflation. Thus, the disclosure requirement is not sufficient to protect consumers from being misled by IRs. These findings motivate us to raise an important question: is there an alternative way to eliminate reviewers' incentives to distort IR ratings? In this section, we examine another type of IR to answer this question.

According to our interviews with active incentivized reviewers, in an IR campaign, firms reach out to reviewers who are likely to post positive and informative reviews as judged by their past IRs. Reviewers agree to post IRs in exchange for rewards from the firm. This suggests that inflated ratings result from compatible incentives between the reward providers (firms) and receivers (reviewers): firms encourage inflated ratings to achieve higher demand for their products, and reviewers write overly positive IRs to obtain greater future rewards. Given this reasoning, we propose that reviewers would have little incentive to inflate their ratings if IRs were commissioned by a third party that benefits from fair (rather than inflated) reviews. Uninflated reviews do not necessarily destroy firms' motivation to participate in IRs because they increase review volume, which is one of the two key word-of-mouth metrics (i.e., valance and volume) having a strong positive impact on sales (e.g., Liu 2006, Gong et al. 2017).

An online platform (e.g., Amazon, eBay, Airbnb, TripAdvisor) is a good candidate for being such a third party. One of the platform's main goals is to facilitate transactions between two parties (i.e., firms and consumers). Thus, the platform benefits from fair reviews for all participating products/services as these reviews allow consumers to make better informed decisions (Dellarocas 2003). Thus, we formally define two types of IR system: firm- and platform-initiated, which differ in who commissions the IRs (thus far, all our reported empirical findings are based on firm-initiated IRs).

In what follows, we provide an empirical analysis of the PIRs using Amazon's Vine program as our empirical context. In the Vine program, Amazon posts a set of free products to its website. From this list of products, selected reviewers (called Vine Voices) pick one or two items to review each month.¹⁶ Reviewers are invited to the program by Amazon, while firms voluntarily participate by providing free products. Reviewers remain anonymous to the firms, however, because Amazon distributes products to reviewers at its discretion. We identify Vine reviews using the "Vine reviews" tag attached by Amazon. For the analyses in this section, we use the *Consumer Reports* data set (introduced in Section 2.2). To exploit within-product variation, we restrict our attention to 432 products with at least one Vine and one verified review.

To examine whether PIRs have inflated ratings, we test for the rating difference between Vine and verified reviews by estimating Equation (1), replacing DIR_{ij} with the PIR dummy (PIR_{ij}), which is one for Vine reviews and zero for verified reviews. The coefficient for PIR_{ij} captures the difference in ratings between verified reviews and PIRs. According to the estimation results reported in column (1) of Table 7, the estimated coefficient of PIR_{ij} is not significant ($\hat{\alpha} = -0.035$, $p > 0.10$). In other words, we find no evidence of rating difference between verified and Vine reviews.

Whereas Vine review ratings are overall uninflated, it is possible that low-quality products may have inflated Vine ratings. To examine whether this may be the case, we estimate Equation (1) by replacing DIR_{ij} with PIR_{ij} and adding an interaction term $PIR_{ij} \times Q_i$, where Q_i is the percentile of product i 's quality score within a category. As reported in column (2) of Table 7, we continue to find that the ratings are not systematically different between platform-initiated IRs and VRs. Importantly, we find no evidence for the moderating role of product quality. This result implies that PIRs are not significantly

Table 7. Difference in Ratings: Platform-Initiated IRs and Verified Reviews

	(1)		(2)	
	Estimate	Standard error	Estimate	Standard error
Platform-initiated IRs	−0.035	0.032	−0.027	0.049
Platform-initiated IRs × Quality			−0.015	0.068
Log reviewer ranking	0.051***	0.004	0.051***	0.004
Log reviewer number of reviews	0.163***	0.010	0.163***	0.010
Time	−1.21E-04***	4.10E-05	−1.21E-04***	4.10E-05
Order	−8.51E-06	1.40E-05	−8.52E-06	1.40E-05
Product-level fixed effects	Yes		Yes	
R ²	0.087		0.087	
N	180,966		180,966	

Notes. Standard errors are clustered at the product level. For this table, we use PIRs and VRs posted to products in the *Consumer Reports* data set. We restrict our attention to products with at least one PIR and one VR to exploit the within-product variation.

***Significantly different from zero at $p < 0.01$.

inflated even for lower quality products and that reviewers of PIRs remain largely unaffected by financial gain from the free products they receive regardless of the product quality.

6. Conclusion

Online reviews are a credible source of information for consumers. To take advantage of this credibility, many firms incentivize reviewers to post reviews for their products, especially in the early stages of the product life cycle given the consequential impact of earlier reviews (Kuksov and Xie 2010, Park et al. 2021). The concern is that, because of financial ties between firms and reviewers, IRs are rarely impartial and, if undisclosed, likely to mislead consumers' purchase decisions. The current FTC guideline recommends that material connections should be disclosed in IRs. In this paper, we assess the efficacy of this recommendation by examining whether disclosure of IRs is effective as a self-discipline mechanism for reviewers and a warning signal to consumers. We also aim to suggest an alternative way to remove the bias in IRs, keeping their volume of information.

Our empirical investigation of DIRs reveals that their ratings are inflated compared with VRs, suggesting that the disclosure cannot discipline reviewers to be strictly impartial in their IRs. Moreover, our analysis shows that inflated ratings of DIRs increase product sales, implying that disclosure cannot help consumers to fully discount the rating inflation. We further find that DIRs propagate one-star reviews, indicating that some consumers are indeed misled by IRs even in the presence of the disclosure. Together, our analyses suggest that disclosure is not an effective measure to prevent distorted reviews or protect consumers from being misled by inflated ratings.

We find, however, that an alternative IR system can restore the impartiality of IRs. Note that the rating distortion arises because inflated ratings align firms' and reviewers' incentives. When the platform provides incentives to reviewers, however, inflated ratings no longer align incentives of the two parties because the platform, as the marketplace benefiting from fair transactions of all products, does not gain from IRs' inflated ratings for any particular product. Hence, switching to platform-initiated IRs can remove rating distortion. Empirically, we find a consistent result: the ratings are not systematically different between Amazon's Vine and verified reviews.

The results of our study have important implications. First, our research calls for consumers' attention to IR disclaimers and their awareness of the differences between incentivized and organic reviews as well as between firm- and platform-initiated IRs. Second, our results suggest that merely requiring a disclosure of an

incentivized review is insufficient. We find that the disclosure can neither keep the review system free of bias nor protect consumers from such potential bias. This calls for stronger platform or government policies than simply requiring disclosure of financial relationships.¹⁷ Third, review platforms should carefully design the review ecosystem, providing specific guidelines for IRs. We show that, even with disclosure, IRs may mislead consumers into suboptimal choices because of overly positive review content. In this process, we also underscore the importance of considering the incentives of self-interested reviewers (see also Kim et al. 2019, Chung et al. 2020 for how reviewer incentives influence their ratings). Platforms may consider adopting platform-initiated IRs, which could alter reviewers' incentives.

A few avenues for further research follow. First, we primarily use Amazon as our empirical context; however, our findings for this platform may not perfectly translate to other online platforms. Thus, it would be worthwhile to extend the study to other contexts (e.g., Fradkin and Holtz 2023). Second, whereas we suggest one alternative (i.e., platform-initiated IRs) to the current policy, our research also calls for more academic research on many other measures that can keep the review ecosystem impartial and protect consumers from the potential bias in IRs. When the results of the current paper were presented to the FTC's division of advertising practices, its consumer protection attorneys indicated that the FTC is preparing a new recommendation related to the IR practice. Academic research can provide useful inputs in this process. We leave these for future research.

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Endnotes

¹ See https://www.ftc.gov/system/files/ftc_gov/pdf/P214504ChairStatementFakeReviewsANPR.pdf (accessed March 18, 2023).

² See <https://www.ftc.gov/business-guidance/resources/featuring-online-customer-reviews-guide-platforms> (accessed March 18, 2023).

³ See BBB National Advertising Division, "5 Tips for Truthful and Transparent Influencer Marketing and Product Reviews" (accessed March 18, 2023).

⁴ Smiley360's disclosure statement provides an example of such encouragement: "Disclosure is an important part of being a Smiley member. Disclosing that you have received a free product means you're following guidelines set by the Federal Trade Commission (we wouldn't want you to get in trouble!) Following these guidelines

will help you to enjoy your Smiley membership and all that it brings.” (<https://smiley.socialmedialink.com/dashboard#/footer/disclosure>, accessed March 8, 2023).

⁵ Past research also investigates fake reviews, which differ substantially from IRs (e.g., see He et al. 2022).

⁶ See FTC Trade Regulation Rule on the Use of Reviews and Endorsements (16 CFR Part 465), and Advance Notice of Proposed Rule-making, Request for Public Comment, Federal Register Vol. 87, No. 215, November 8, 2022.

⁷ For Amazon’s announcement, see <https://www.aboutamazon.com/news/innovation-at-amazon/update-on-customer-reviews> (accessed February 21, 2023).

⁸ In our study, we identify DIRs using a set of disclaimers as well as their variations, such as “provided a free sample,” “received a free sample,” “received this for free,” “received this product for free,” “received this at a discount,” “received a sample,” and “received this product at a discount.”

⁹ For more details of the *Consumer Reports* data set construction, see Online Section A1. Also, note that this data set includes a distinct set of products compared with our main data set.

¹⁰ It is useful to note that we do not find evidence that the likelihood of DIR posting significantly differs across quality levels (see Online Section A3 for details). Thus, our results are not driven by the concentration of DIRs on products of a certain quality level.

¹¹ After a significant amount of time, we no longer observe the second group of products and find that all DIRs had been deleted as of December 2018.

¹² Note that both average ratings and log number of reviews are calculated from all reviews that are not DIRs. Including DIRs into the calculation biases the estimation of the DIR removal effect by making these control variables partially capture the effect of DIR removal.

¹³ One may wonder whether the decrease in the number of one-star reviews is driven by the sales decrease from DIR removal. However, we find qualitatively the same results ($\hat{\lambda} = 0.031$, $p < 0.05$) even after controlling for log sale rank in Equation (3).

¹⁴ For Amazon’s announcement, see <https://blog.aboutamazon.co.uk/shopping-and-entertainment/update-on-customer-reviews> (accessed June 4, 2023).

¹⁵ For details of the construction of the Amazon UK data set, see Online Section A1.

¹⁶ For more details, see <https://www.npr.org/templates/transcript/transcript.php?storyId=247833514> (accessed February 21, 2023).

¹⁷ Although firm-initiated IRs have been banned from Amazon, they are still observed in many other platforms such as BestBuy, Home Depot, Lowe’s, Target, and Walmart, as of June 2023.

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